

Principles of 3D computer vision

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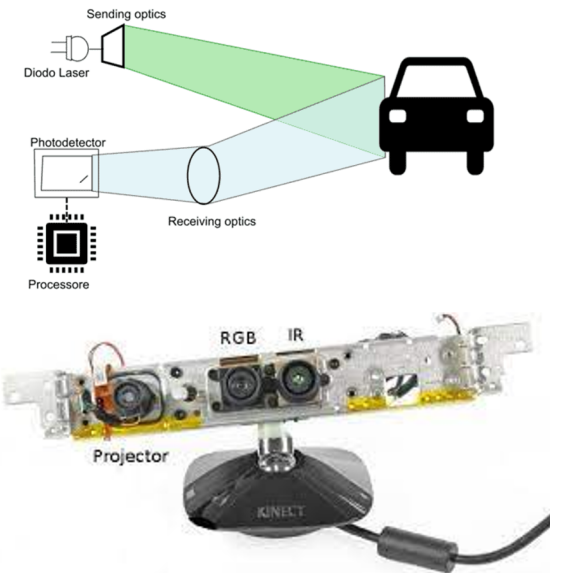


Introduction

Active 3D Sensors

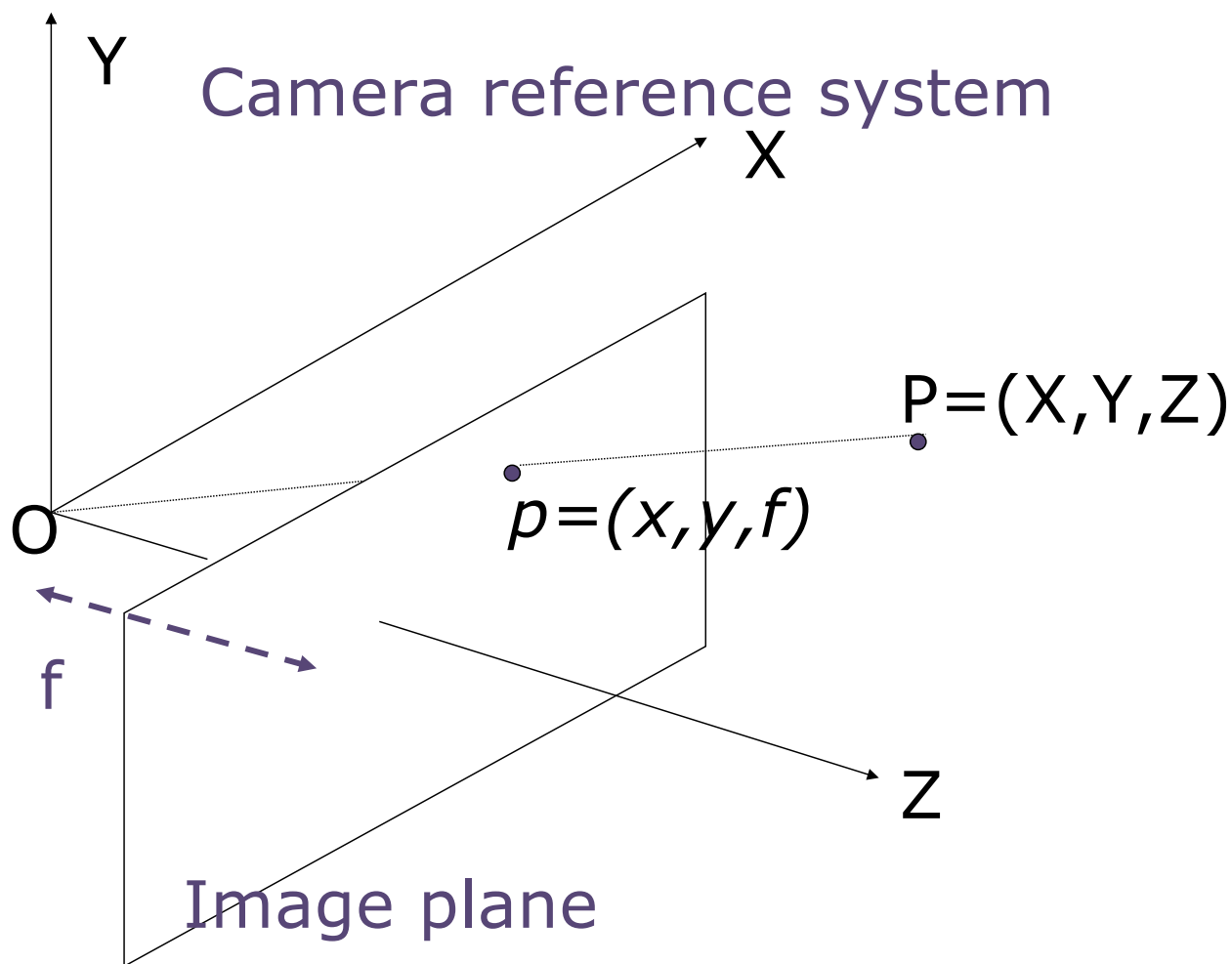
Time-of-Flight active sensors to measure distance

- LIDAR (laser based, short to medium range)
- IR or RGBD (IR based, short range)
- RADAR (RF based, medium to long range)

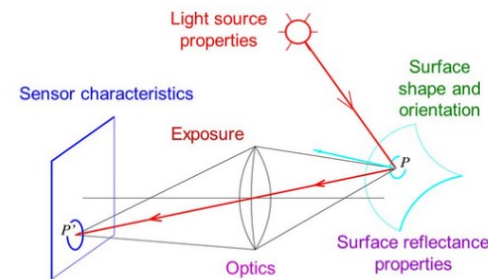


RGB camera sensors

The geometry of image formation



f focal length

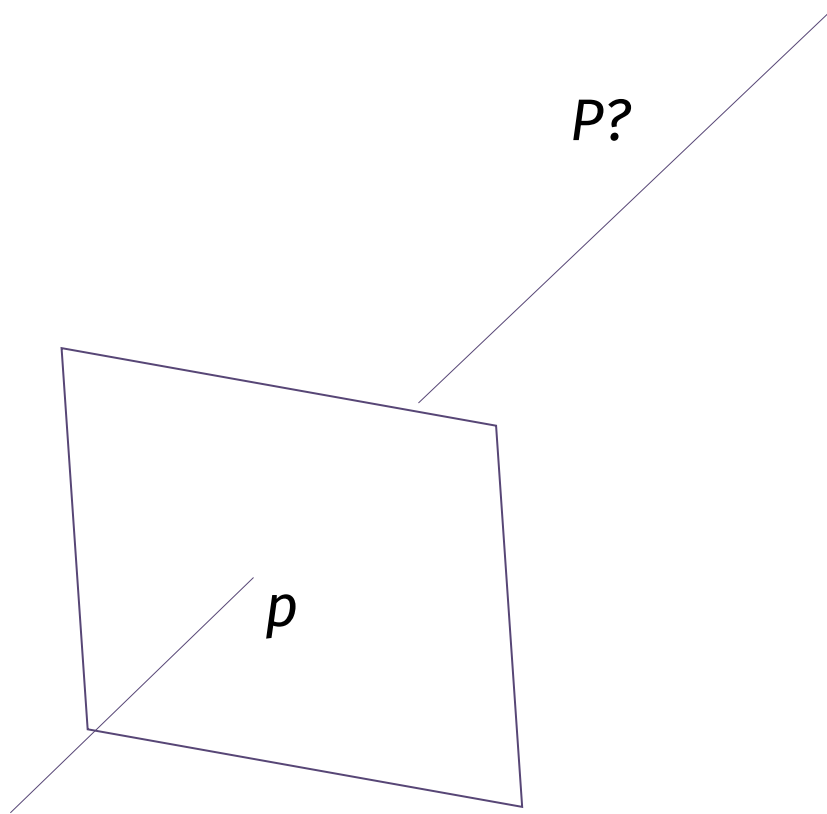


Scale Ambiguity

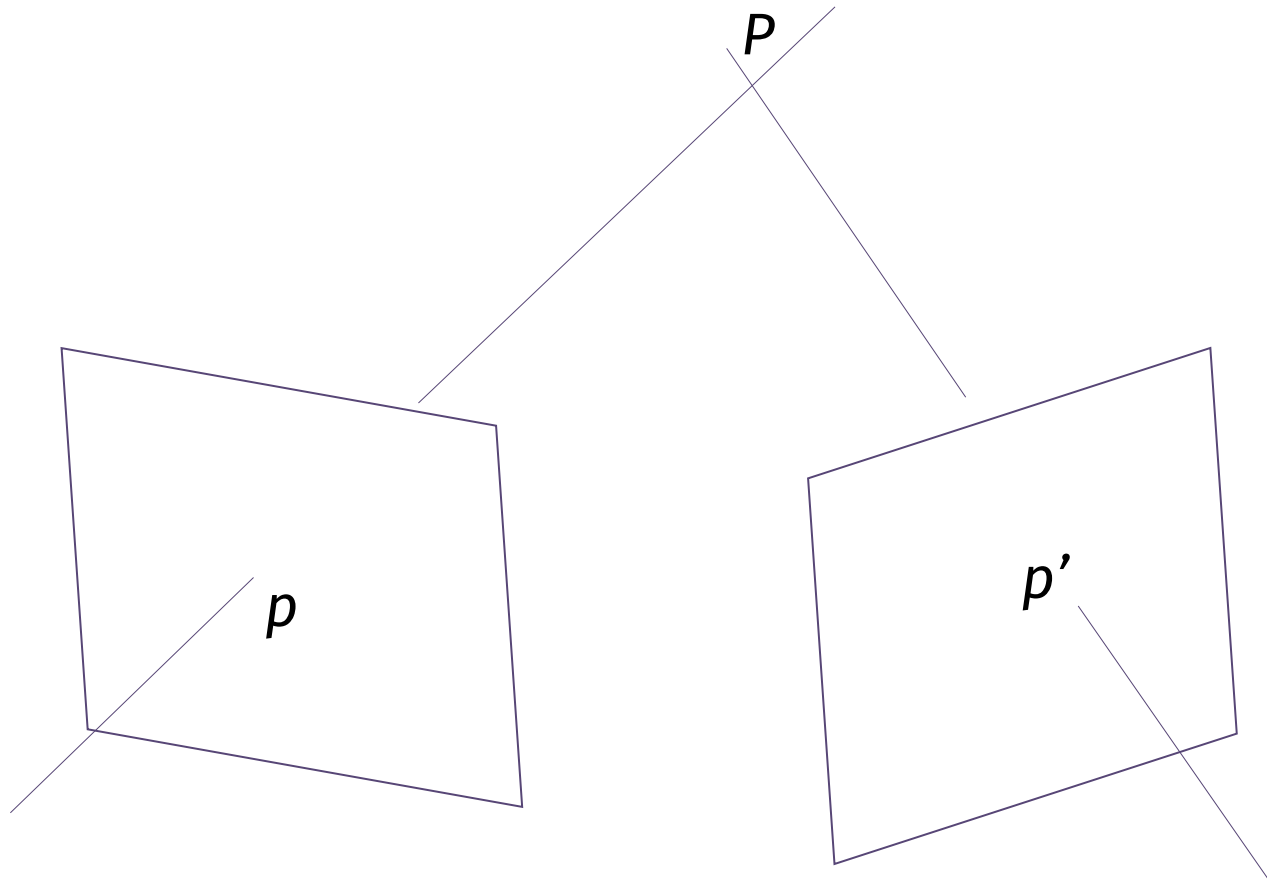
- 3D Information is lost during the projection to the image plane (or the retina)
- It is not possible (without a proper prior knowledge) to distinguish small objects from far objects



The perspective projection taking place when an image is acquired is a “lossy” non invertible transformation



With a second view...





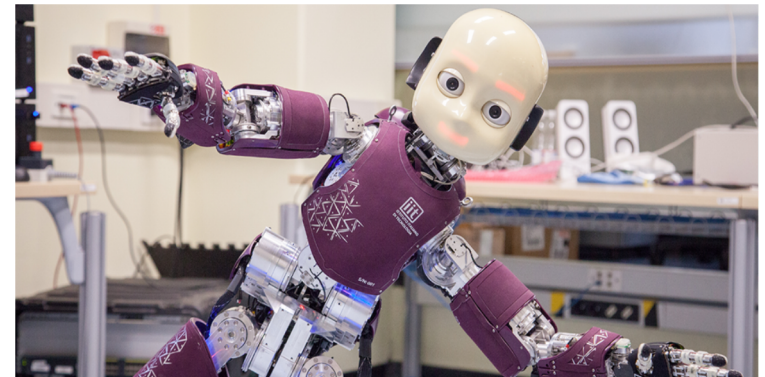
Stereopsis

Introduction

we refer to **stereo vision** as the problem of inferring 3D information (structure and distances) from two or more images taken from different viewpoints

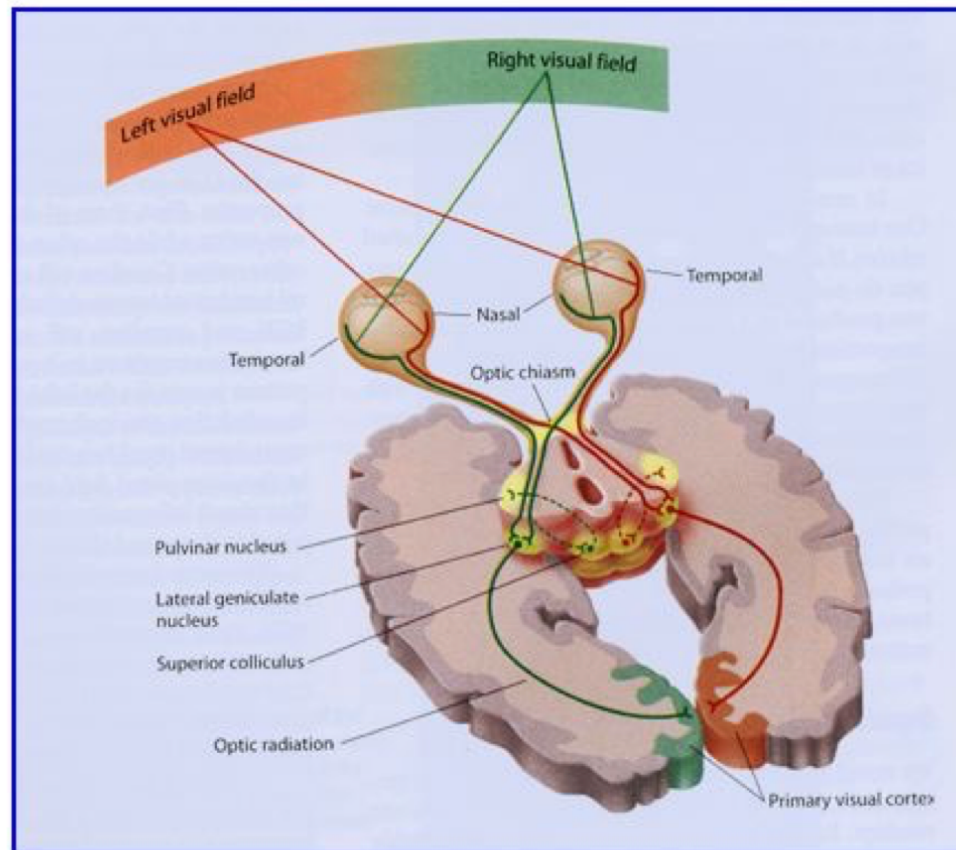
we consider an acquisition system with 2 cameras

- “explicit” system: a stereo rig
- “implicit”: one moving camera



Stereopsis: perception of depth

Our 3D perception of the world is due to the interpretation that the brain gives of the computed difference in retinal position, called *disparity* between corresponding items



A simple stereo system

Relationship between depth and disparity

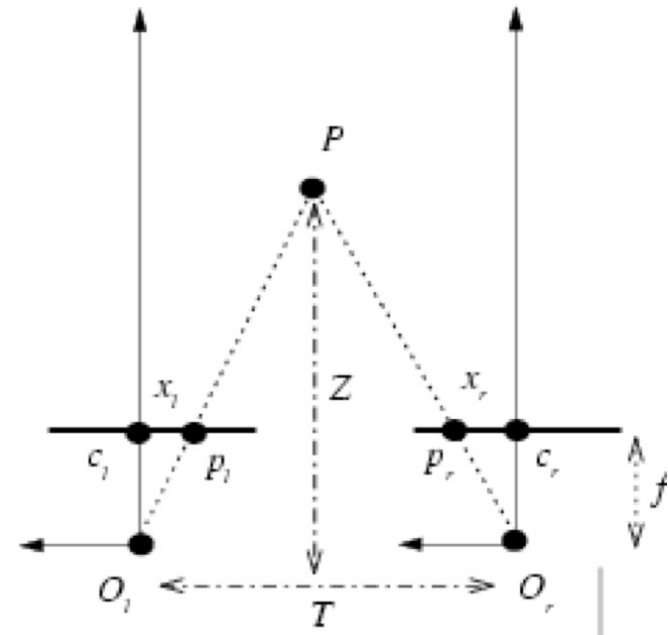
We can recover depth information

$$Z = \frac{fT}{x_r - x_l} = \frac{fT}{d}$$

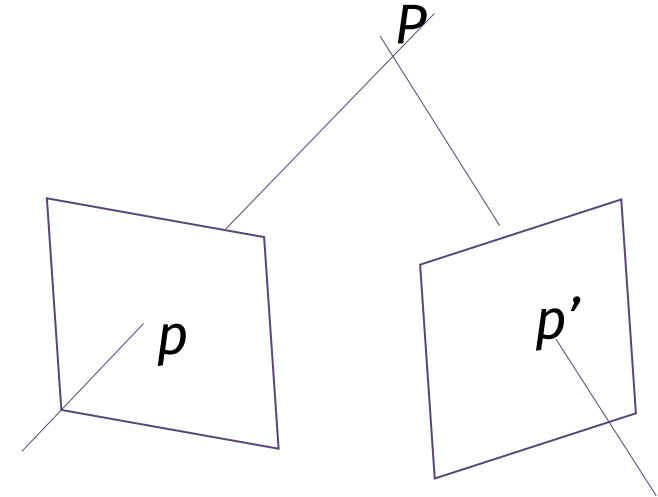
Diagram illustrating the relationship between depth (Z) and disparity (d). The equation shows that depth is the product of focal length (f) and baseline (T) divided by disparity (d). The term d is circled, and arrows point from the labels "focal length", "baseline", and "disparity" to their respective terms in the equation.

Depth is inversely proportional to disparity

Special case:
fixation point at infinity



The problems of stereopsis



stereo covers two main problems:

- finding **correspondences** between image pairs (p, p')
- **reconstructing** the 3D position of a point P given its corresponding projections on the images

step 1: produces 2.5 disparity maps (or depth maps)

step 2: produces a 3D point cloud



Disparity

Correspondence problem

the correspondence problem involves two decisions:

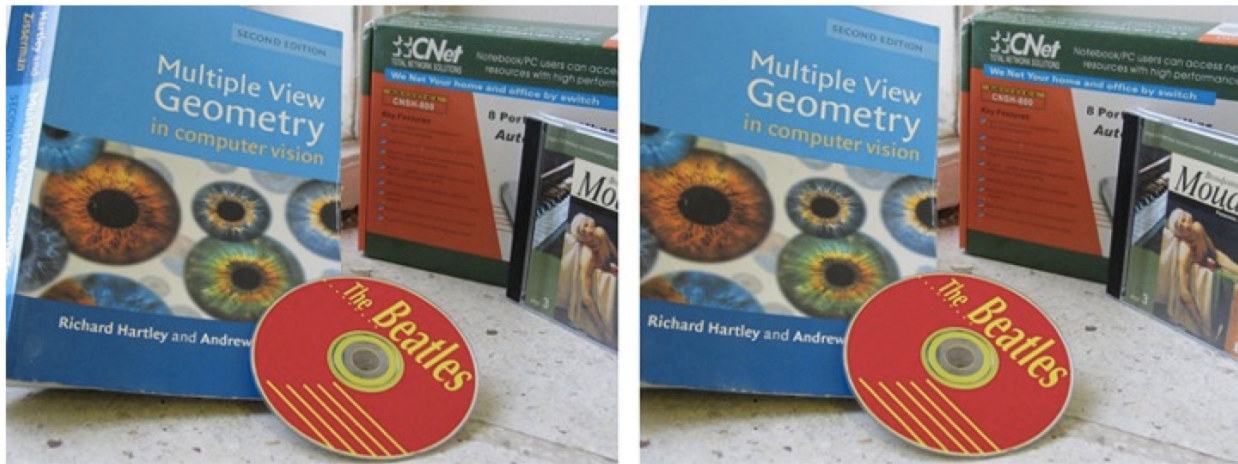
1. which image elements to match
2. which feature description + similarity measure to adopt

(1). can be solved in two ways:

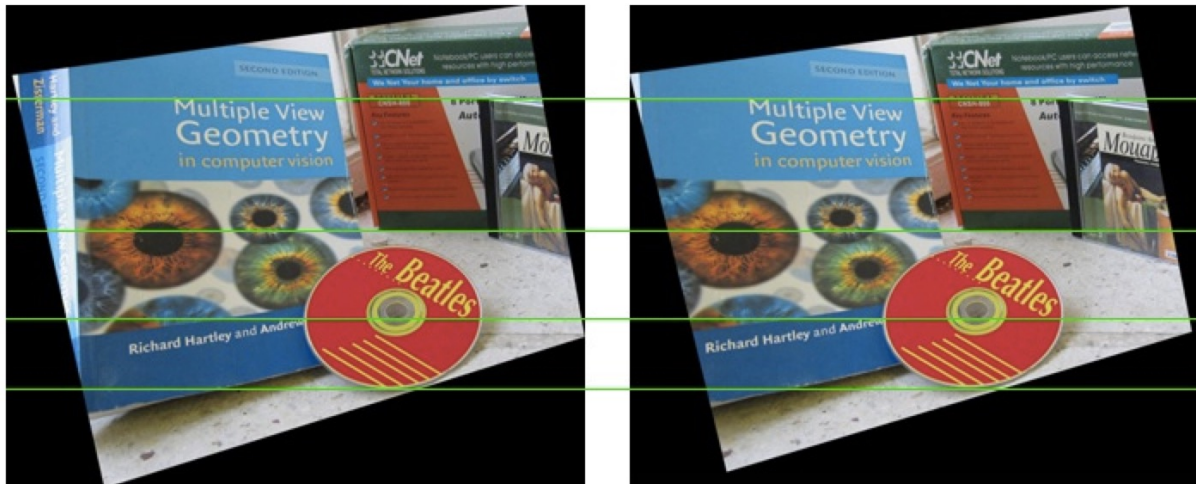
- use all pixels in the image (obtaining *dense correspondences or disparity maps*)
- use subsets of pixels meeting some requirements (obtaining *sparse correspondences*)

Computing disparity maps

Two words on input images



A stereo pair



A rectified stereo pair

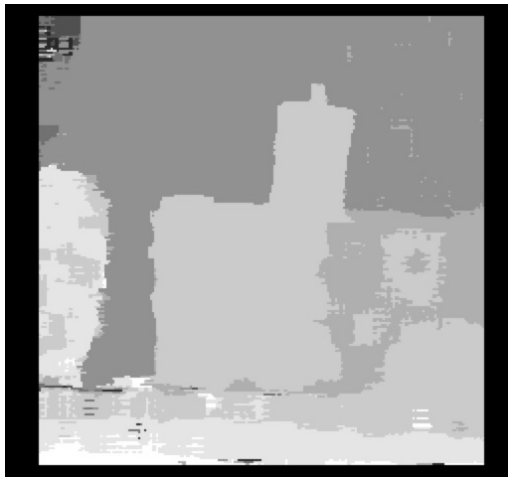
Equiv to fixation point at infinity

Dense correspondences

we consider the so-called *correlation methods* applied to image patches (neighbourhoods of pixels)

we assume we have two *rectified images*, where conjugate points lie on corresponding scanlines of the image (“rows”)

our goal is to obtain a **disparity map** giving the relative displacement for each pixel



assuming a fixation point at infinity
disparity is proportional to the inverse
of the distance

in a standard color coding bright areas
correspond to high disparities (closer
objects)

Dense correspondences: algorithm sketch

input:

- a stereo pair of rectified images l and r
- size of a correlation window W
- a search range $[d_{min}, d_{max}]$

for each pixel p of (i,j) coordinates in l

- for each disparity d in the search range

- estimate the similarity $c(d) = \phi(N1(i, j), N2(i, j + d))$

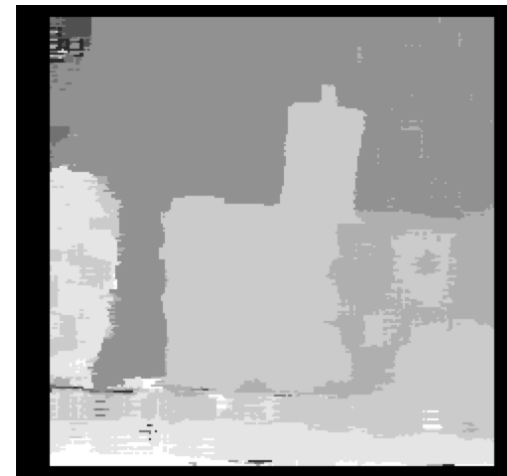
*For instance SSD or NCC
(see image matching class)*

- the disparity of the pixel is $\bar{d} = \operatorname{argmax}_{d \in [d_{min}, d_{max}]} \{c(d)\}$

Dense correspondences: left-right consistency

- correspondences are made more difficult by occlusions (points with no counterpart on the other image)
- let us compute
 - D_{lr} : disparity map from l_l to l_r
 - D_{rl} : disparity map from l_r to l_l
- then $D(i,j)=d$ iff $D_{lr}(i,j) = -D_{rl}(i,j+d) = d$

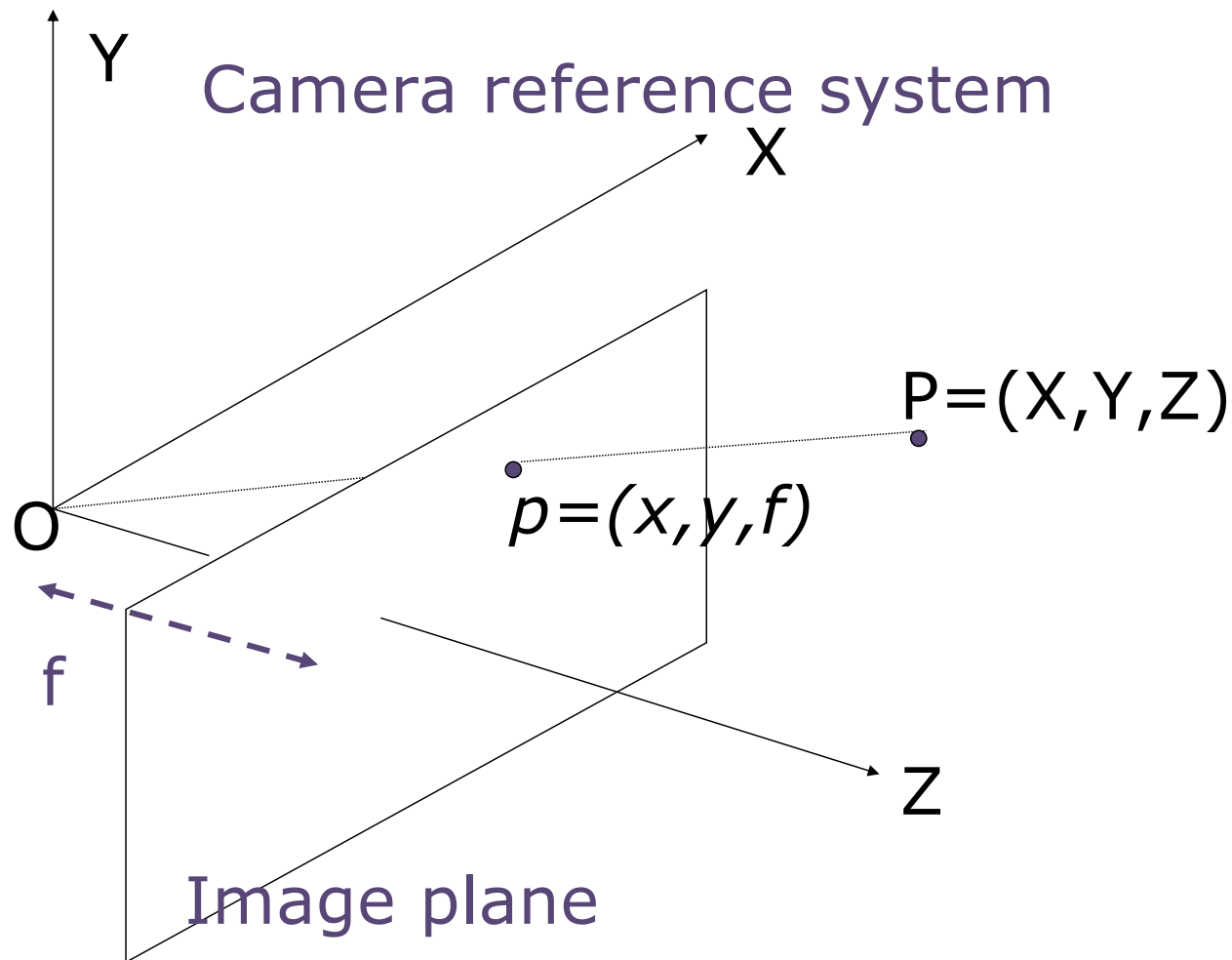
Examples





3D reconstruction

Geometry of image formation: perspective or pin-hole model



$$x = f \frac{X}{Z}$$
$$y = f \frac{Y}{Z}$$

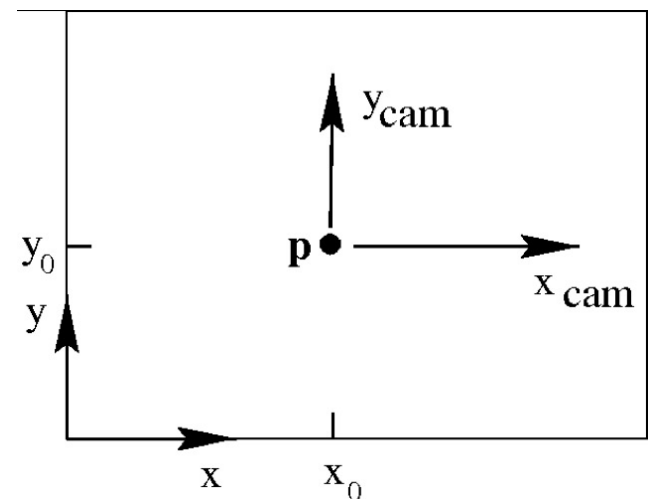
f focal length

Parameters of a stereo system

How to relate points in the world with pixels in the image

Intrinsic/internal parameters

- characterize the mapping of an image point from camera to pixel coordinates in each camera
- They include focal length, the mm-pixel change of coordinates (scaling), the position of the principal point (translation)



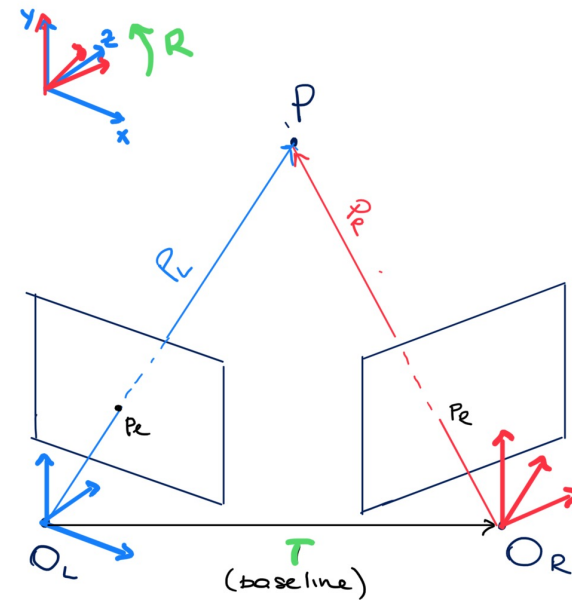
Parameters of a stereo system

How to relate points in the world with pixels in the image

Extrinsic/external parameters

- describe the relative position and orientation of the two cameras ($R, \mathbf{T}$)

$$\mathbf{P}_r = R(\mathbf{P}_l - \mathbf{T})$$



System calibration: sketch

Internal parameters can be known a priori or estimated by camera calibration procedures

External parameters can be obtained starting from point correspondences

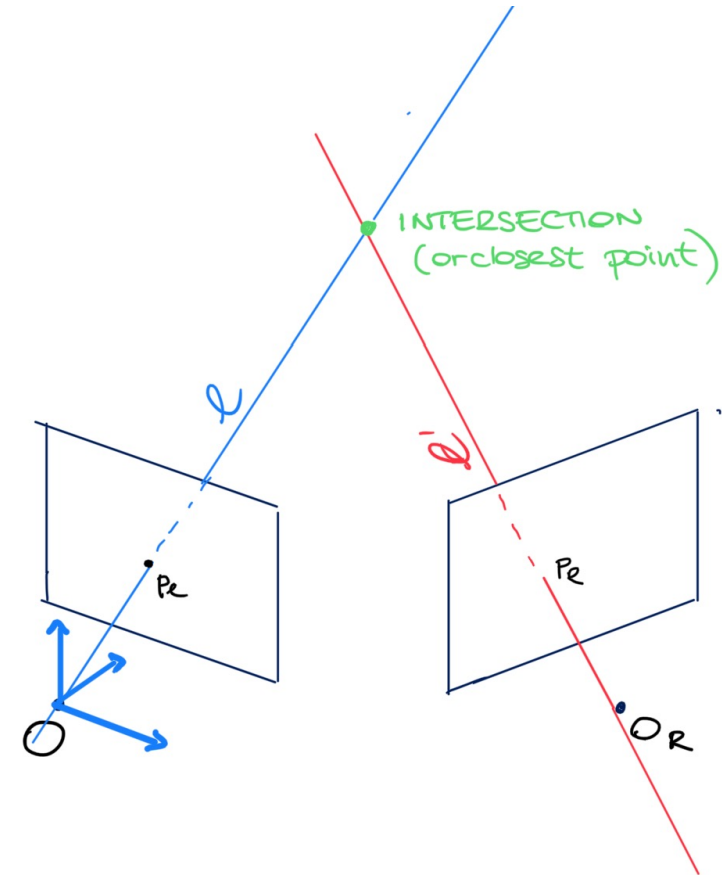
2D to 3D: points triangulation (sketch)

Assuming we know the internal and external parameters

Given a pair of corresponding points in 2D (metric coordinates on a common reference frame) (p_l, p_r)

Estimate the equation of the two projection rays ℓ, ℓ'

Compute the intersection (or the closest point wrt both lines) it will be the 3D reconstruction P





quick detour on geometry background

Homogeneous coordinates

Points on a plane

A point on a plane $(x, y) \in \mathbb{R}^2$

Lines

Equation of a line on a plane $ax + by + c = 0$

We may also represent it as $(a, b, c)^\top$

Notice that $(ka, kb, kc)^\top$ $k \neq 0$ it is the same line (they belong to the same **equivalence class**)

This equivalence class is called a homogeneous vector

Homogeneous coordinates

Projective plane

The projective plane (or 2D projective space) is formed by any vector $(a, b, c)^\top$ representative of an equivalence class:

$$\mathbb{R}^3 - (0, 0, 0)$$

Homogeneous coordinates

A point and a line

A point $(x, y)^\top$ lies on a line $\mathbf{l} = (a, b, c)^\top$ IFF

$$ax + by + c = 0$$

If we define $\mathbf{x} = (x, y, 1)^\top$ then

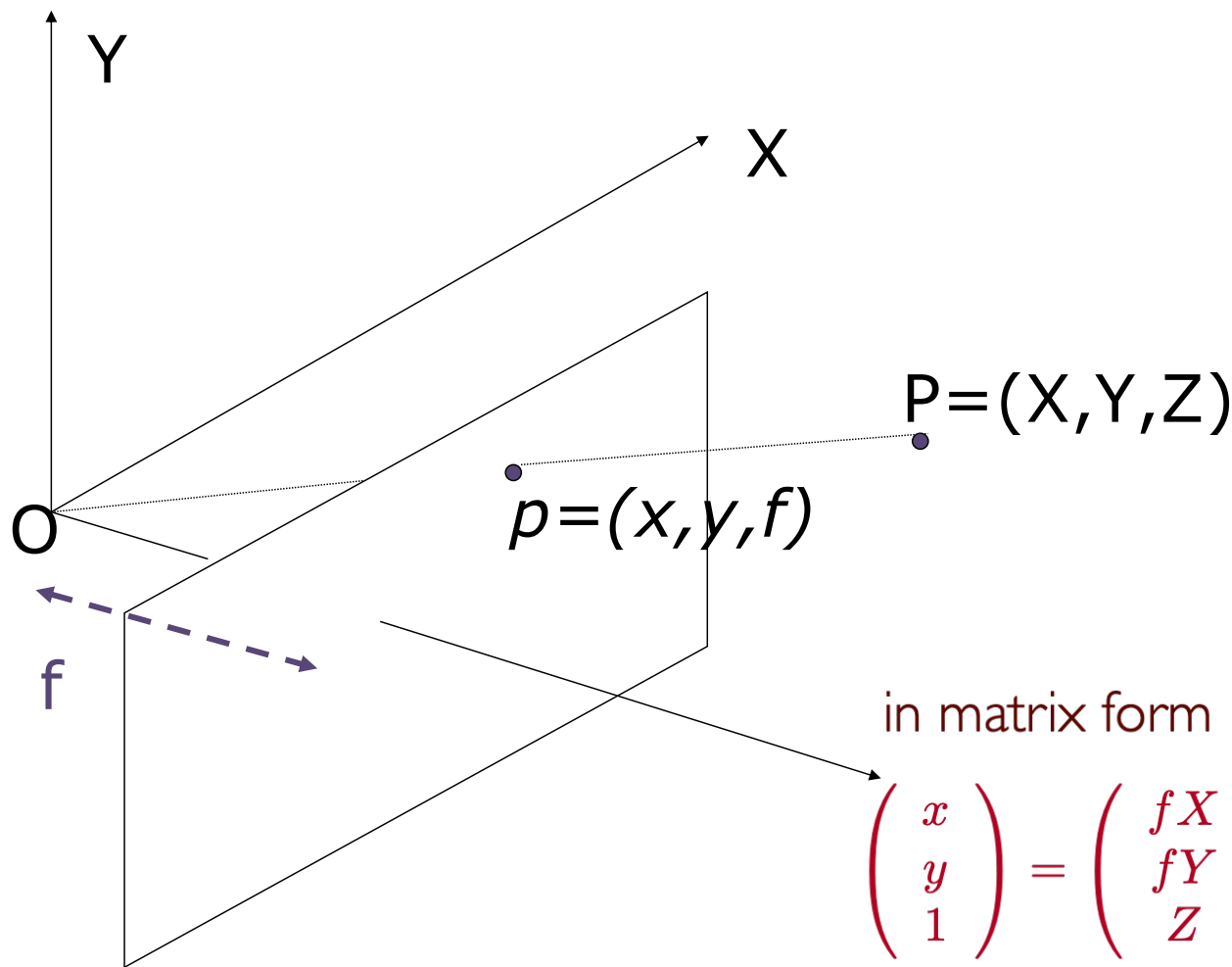
$$\mathbf{x}^\top \mathbf{l} = 0$$

A point on the Euclidean plane may be represented as a point in the projective plane (as a 3D vector with a final 1).

Notice that here again points define equivalence classes

Perspective model

in homogeneous coordinates becomes a linear transformation



$$x = f \frac{X}{Z}$$
$$y = f \frac{Y}{Z}$$

in matrix form (homogeneous coordinates)

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Parameters of a stereo system

How to relate points in the world with pixels in the image

Intrinsic/internal parameters

- characterize the mapping of an image point from camera to pixel coordinates in each camera
- (translation and scaling)

$$K = \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$$

all entries in pixel coordinates

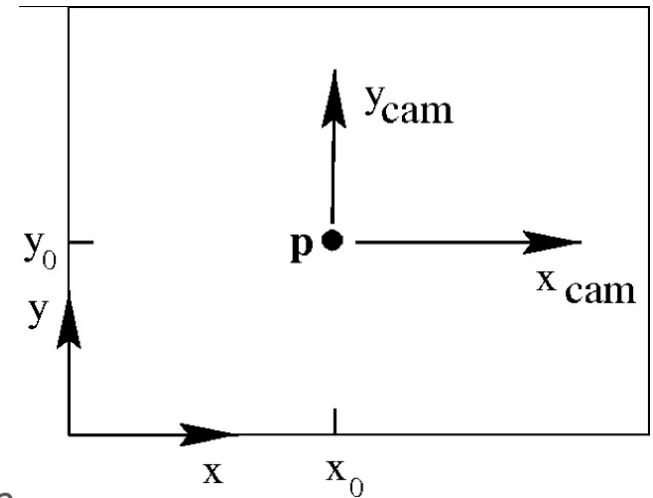
$$\alpha_x = f m_x$$

$$\alpha_y = f m_y$$

$$x_0 = m_x p_x$$

$$y_0 = m_y p_y$$

number of pixels per unit distance

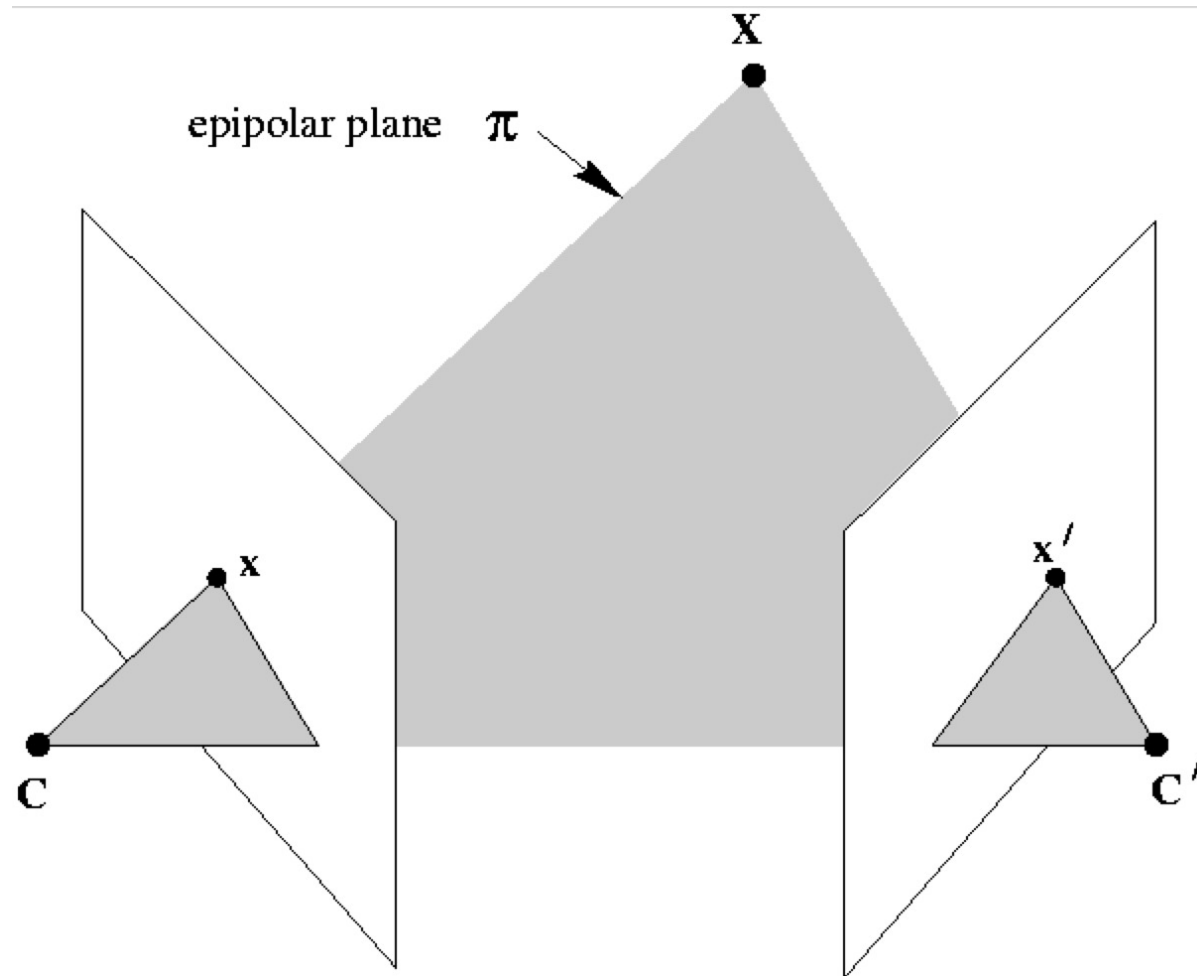


$$\mathbf{p}_{pix} = K M \mathbf{P}$$

Epipolar geometry

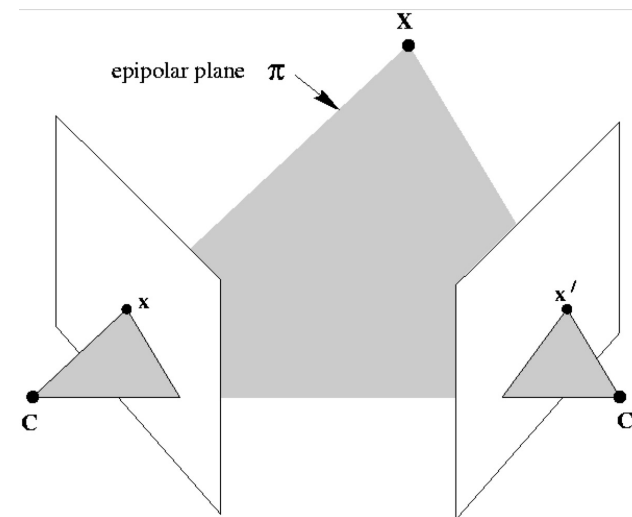
Geometry of a stereo system

Epipolar geometry



Geometry of a stereo system

Epipolar constraints



let us consider x How is x' constrained?

x' lies on the line l' intersection between the epipolar plane and the (right) image plane

l' is the projection on the (right) image plane of the ray passing through C and x

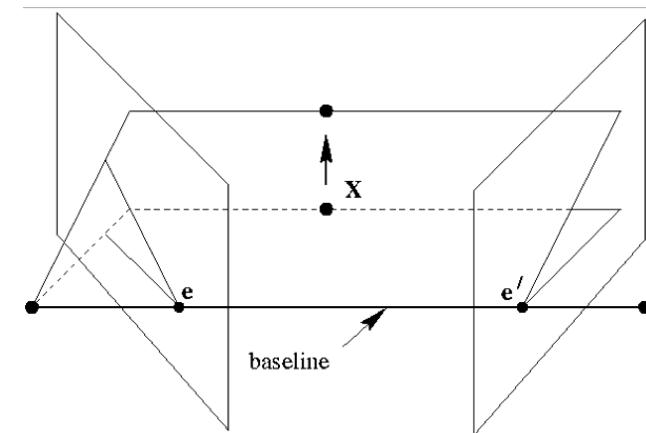
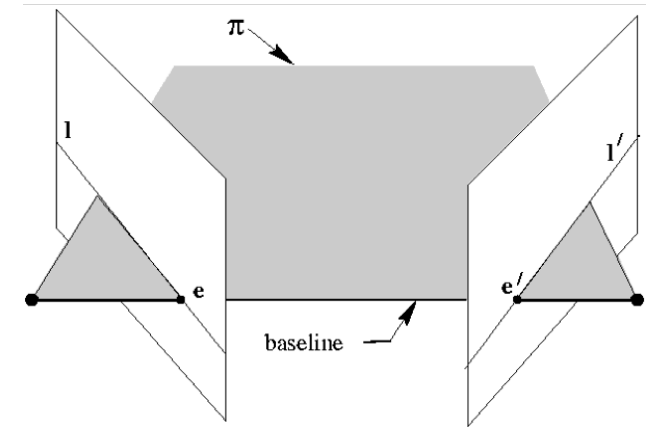
In practice, when we look for the corresponding point x' we can limit our search to a line, l'

from a 2D to a 1D problem!

Geometry of a stereo system

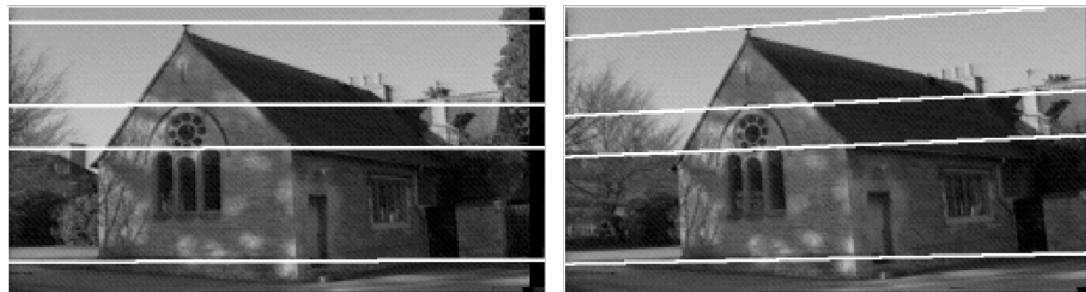
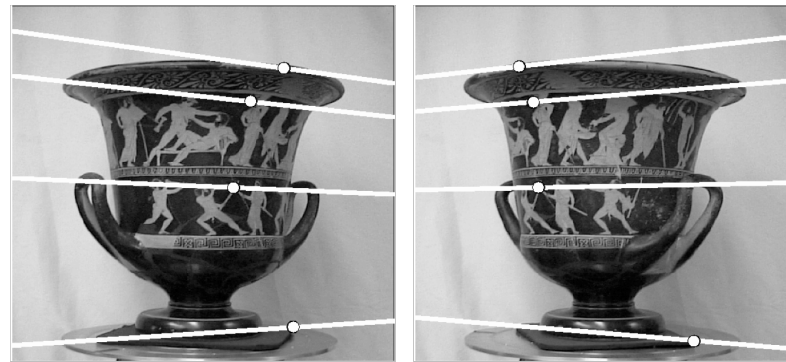
Terminology

- **epipole:** the intersection of the baseline with the 2 image planes. Also: the projection of one camera centre to the other image plane
- **epipolar plane:** a plane containing the baseline and passing through a 3D point $\mathbf{X}$. It is a *one parameter family (stencil)* of planes
- **epipolar line:** the intersection of an epipolar plane with one image plane. All epipolar lines contain the epipole.

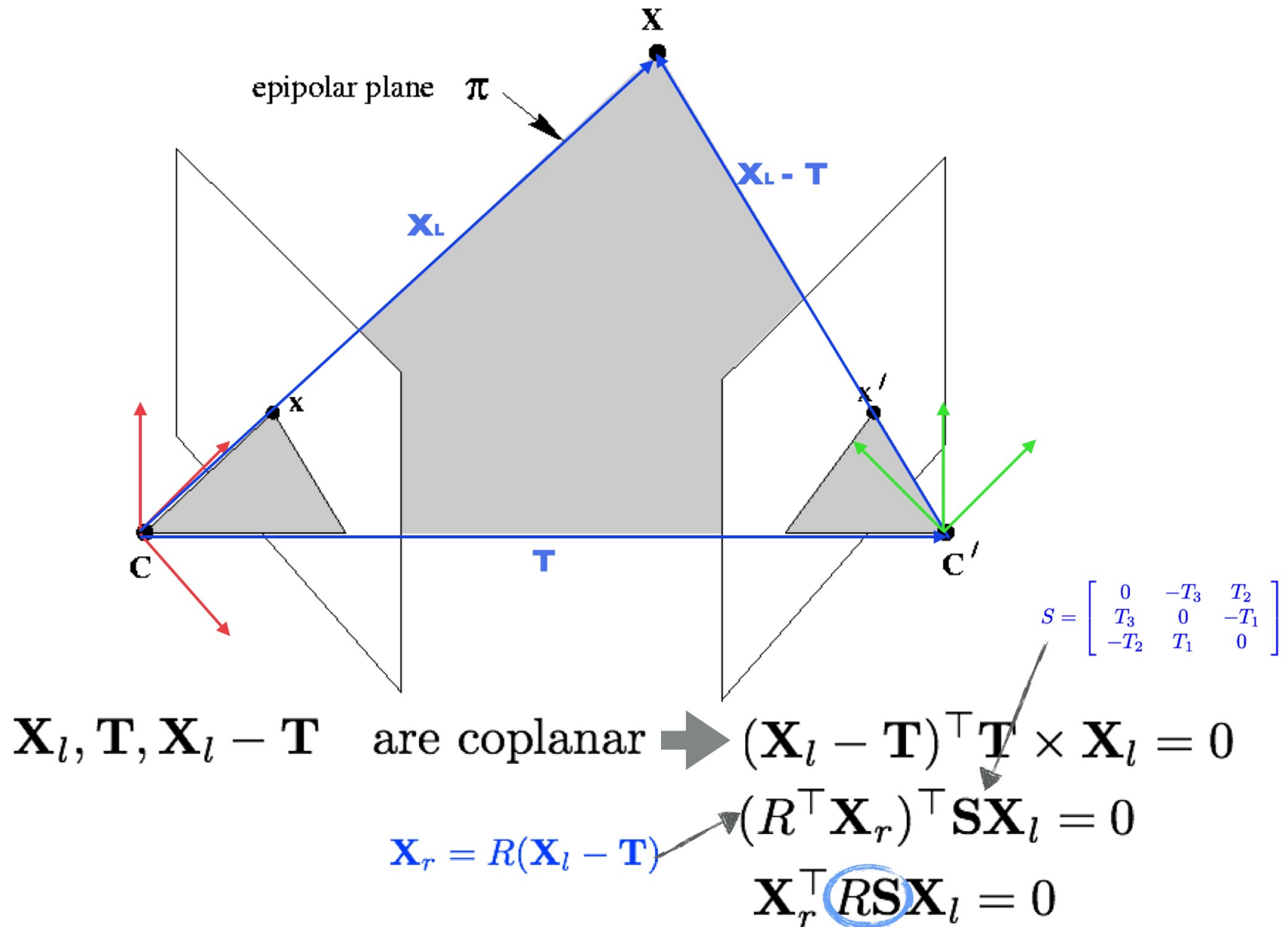


Geometry of a stereo system

Epipolar lines



Geometry of a stereo system



Geometry of a stereo system

Essential matrix

- The essential matrix is a 3x3 matrix $E=SR$ $\mathbf{X}_r^\top E \mathbf{X}_l = 0$
- by projecting the point on the two image planes we obtain the equation which is *satisfied by each (x_m, x_m') pair of corresponding points in mm coordinates*

$$\mathbf{x}_m'^\top E \mathbf{x}_m = 0$$

- E has rank 2, since S has rank 2 and R is full rank
- E has 5 d.o.f. : 3 from R 3 from $\mathbf{T}$ but with a scale ambiguity $3+3-1=5$

Geometry of a stereo system

Fundamental matrix

we now derive an equivalent equation relating points in **pixel coordinates**:

$$\mathbf{x}_m = K_l^{-1} \mathbf{x}$$

$$\mathbf{x}'_m = K_r^{-1} \mathbf{x}'$$

•Then

$$\mathbf{x}'^{\top} \underbrace{K_r^{-T} E K_l^{-1}}_F \mathbf{x} = 0$$

•The fundamental matrix satisfies the equation

$$\mathbf{x}'^{\top} F \mathbf{x} = 0$$

for each pair of $(\mathbf{x}, \mathbf{x}')$ of corresponding points in pixel coordinates

Geometry of a stereo system

Fundamental matrix

Suppose we have two images acquired by cameras with no coinciding centres.

$$\mathbf{x}'^{\top} F \mathbf{x} = 0$$

Then the fundamental matrix F is the unique 3×3 rank 2 matrix so that, for each corresponding pair $(\mathbf{x}, \mathbf{x}')$

Fundamental (and essential) matrix is a map between points and (epipolar) lines

$$\mathbf{l}' = F \mathbf{x}$$

Geometry of a stereo system

8 points algorithm (sketch)

$$\mathbf{x} = (x, y, 1)^\top \quad \mathbf{x}' = (x', y', 1)^\top$$

we may obtain a linear algorithm from 8 points correspondences (notice this is not the minimal problem)

$$\begin{bmatrix} x_1'x_1 & x_1'y_1 & x_1' & y_1'x_1 & y_1'y_1 & y_1' & x_1 & y_1 & 1 \\ x_2'x_2 & x_2'y_2 & x_2' & y_2'x_2 & y_2'y_2 & y_2' & x_2 & y_2 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_n'x_n & x_n'y_n & x_n' & y_n'x_n & y_n'y_n & y_n' & x_n & y_n & 1 \end{bmatrix} \begin{pmatrix} f_{11} \\ f_{12} \\ \vdots \\ f_{33} \end{pmatrix} = \mathbf{0}$$

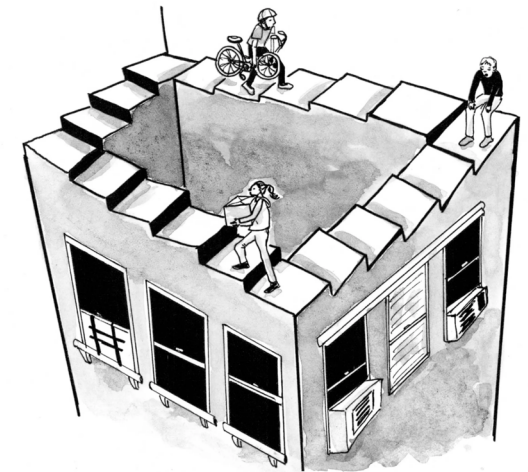
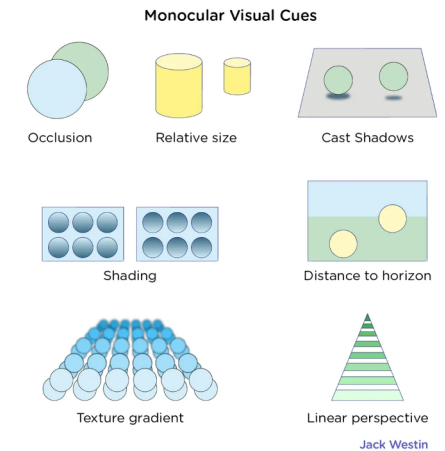
It is a homogeneous system $Af=0$, overdetermined, which may be solved with SVD



Deep learning and 3D vision

It is not all about geometry

- Stereopsis in humans (given the narrow baseline) works primarily in short range (up to 10 meters)
- Beyond this range, the brain uses other cues
- Humans can still infer depth from single images
- This provides a biological justification to the monocular deep learning approaches that have been proposed in the last years



Monocular Vision and Deep Learning

- Many recent works uses deep ConvNets to learn monocular depth directly from RGB images
- Not only depth
- We do not have access to very large general purpose datasets for 3D tasks :
 - Expensive to acquire and annotate images
 - Using the limited amount of available annotations results in poor generalization (despite performing well on benchmark data)
 - Research is focusing on unsupervised or self-supervised approaches

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